

Joanne Hoar

Technology Educator, Software Developer, Game Developer

 [linkedin.com/in/joanne-hoar](https://www.linkedin.com/in/joanne-hoar) —  [xcodegirl](#)

Summary

Canadian Software Developer with 20 years of experience in video game programming, scientific modelling and cross-platform development. Expertise in multiple programming languages, game engines and SDKs. Proven track record in leading teams, mentoring developers, impressing customers and building corporate culture. Strong background in C++, Unreal Engine, iOS and Android using agile methodologies and CI/CD. Passion for algorithmic correctness, mathematics, artificial intelligence and nature.

Experience

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- Technology Instructor**, *Robertson College (contract)* Jan 2025 – present
C#, .NET, ASP .NET Core, Java, JavaScript, React, Git, GitHub, Azure, SQL
Instructed 20+ course offerings across Full Stack Development and Data Analytics diploma programs. Delivered live coding sessions, maintained open-source course materials on GitHub, and achieved an average student satisfaction rating of 4.5/5.0.
- Senior Software Developer**, *bitHeads Inc.* 2022 – 2025
Unreal Engine, C++, Objective-C, Swift, iOS, Android Native, Java, Kotlin, PlayStation, Xbox, Nintendo Switch, Node.js, Groovy, C#, Shell Script
Developed and maintained cross-platform client SDKs for a backend-as-a-service platform used by 50+ published games across 12 platforms including PlayStation, Xbox, Nintendo Switch, iOS, and Android. Engineered CI/CD and TDD pipelines and managed customer relationships with AAA and Indie studios.
- Co-founder/ Lead Developer**, *Silicon Hanna Inc. (self-employed)* 2011 – 2022
Unreal Engine, Objective-C / Swift, iOS, Android Native, Kotlin, Augmented Reality, MacOS, Windows
Designed, coded and published eleven mobile games for iOS and Android, from concept and game design through to commercial app-store release. Experimented with cutting-edge device technologies including ARKit and other emerging SDKs.
- Tools Programmer**, *Fekete Associates Inc. (S&P Global)* 2008 – 2011
C++, MFC, Windows, Perforce, Incredibuild
Developed tools that used mathematical modelling to maximize production and improve efficiency for major oil & gas corporations. Led research on a new system that reduced build times by 40%.
- Intermediate Software Engineer**, *BJ Pipeline Inspection Services Inc. (Baker Hughes)* 2005 – 2008
C++, C, OpenGL, PHP, SQL, Neural Networks, Borland, Windows, Linux
Visualised in-line device data with OpenGL graphics. Implemented neural network algorithms for anomaly detection. Led development of custom web-based GIS reporting software. Improved low-level programming algorithms for data processing.
- Co-founder / App Developer / Instructor**, *SuRJE Software Solutions Inc. (self-employed, contract)* 2005 – 2008
Objective-C, C++, OpenGL, Chipmunk, PHP, JavaScript, Firebase, Google Maps, iOS, Linux
Created and published physics-based games for iOS. Handled technical writing, web development, Google Maps and Firebase integration and server administration. Developed course curriculum, instructed C++ programming and assessed student work with the Department of Continuing Education.
- Software Researcher / Lead Developer**, *PlayStarMusic Corp.* 2005 – 2006
C++, PHP, Javascript, .NET, Windows, Linux
Led development and system administration for an iTunes-like music store. Automated DRM encoding and video streaming. Developed a prototype for a Google-competitive search engine.
- Junior Software Engineer**, *Codefast.com* 2004 – 2005
C++, Qt, Wx, Shell script, Windows, Unix
Ported applications to operate under a new GUI library. Developed user interfaces for software lifecycle automation.
- Web Developer / VR Artist**, *Nickle Arts Museum, University of Calgary (contract)* 2002 – 2004
JavaScript, C++, OpenGL, PHP, HTML, CSS, 3D Scanning, VR
Designed and developed an interactive online catalogue for the Gerald Hushlak art exhibit with JavaScript-based features. Contributed original generative art from graduate research. Collaborated on a pioneering VR installation using 3D scanning and CAVE immersion technology.
- Research Assistant**, *VLab Computer Graphics Group / Artificial Intelligence Research Hub* 2001 – 2004
C++, OpenGL, PHP, MySQL, Qt, Solaris, Unix, Linux

Developed graphics software tools using fractal mathematics and biological modelling. Managed network setup and distributed computing infrastructure. Published five research papers and served as teaching assistant across three courses.

Education

Master's Degree, Computer Science , University of Calgary	2002 – 2004
<i>Thesis: Agent-based Development of Natural Transportation Networks</i>	
Bachelor's Degree, Computer Science , University of Calgary	1998 – 2001

Honours & Awards

Departmental Nomination: Governor General's Gold Medal, Faculty of Graduate Studies	2005
Postgraduate Scholarship NSERC D3, Natural Sciences and Engineering Research Council of Canada	2005
Graduate Student Scholarship, Alberta Learning	2004
Best Student Paper Award (Co-author), ACAL	2003
CPSC Departmental Research Award, University of Calgary	2003

Portfolio

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Published Games

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- **Ragin' Bull** – iOS, Android (2020)
 - **AR Picture Portal** – iOS (2018)
 - **Fragmented Reality** – iOS (2018)
 - **IceCream AR** – iOS (2018)
 - **Jewel Slide** – iOS, Android (2017, 2019)
 - **Word Slide & Find** – iOS (2017)
 - **A Word Theme** – iOS, Android (2017, 2019)
 - **CryptoQuip** – iOS, Android (2011, 2019)
 - **Rafter's Loot** – iOS (2011)
 - **Star Chase** – iOS (2008)
 - **Bitsy** – iOS (2008)